Ji et al. 2020 · Cutaneous Squamous Cell Carcinoma

GO / NO-GO GATES

Gate 1 — Integration & continuity **GO**

Harmony (theta=5) merges 8 samples into one connected kNN component; continuous, no sample islands.

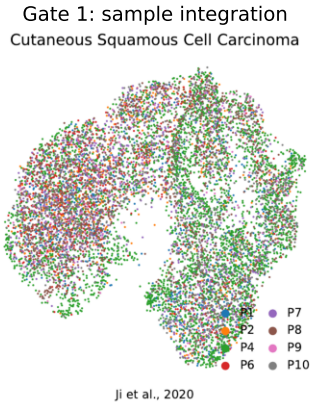
Gate 2 — Module geography **GO (branching)**

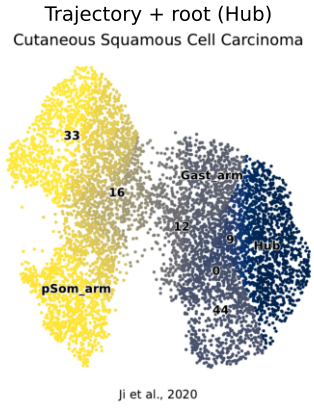
Gast/pSom-high hub diverges into a Gast-arm and a pSom-arm; Crest a terminal readout toward the tips.

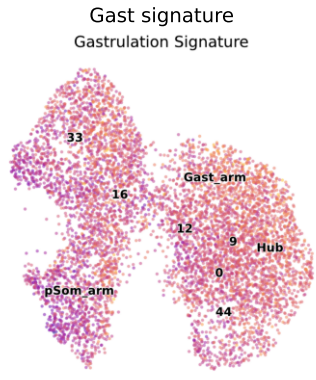
Gate 3 — Hub cross-repression **GO**

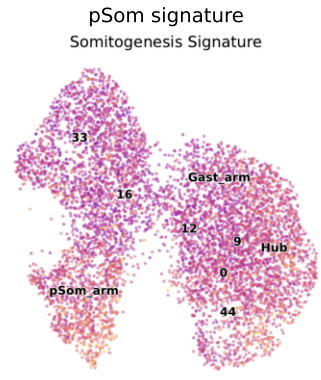
Hub Gast<->pSom module r = -0.37 (perm p = 0.007), competitive; ≈ Puram -0.49 / Filbin -0.34 anchors.

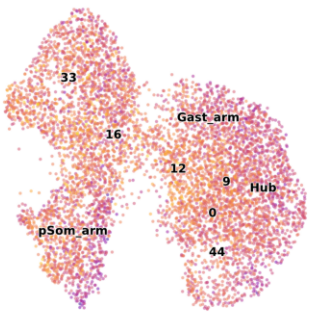
STEP FIGURES

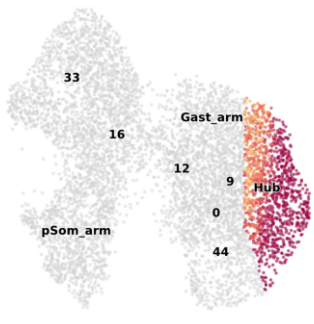


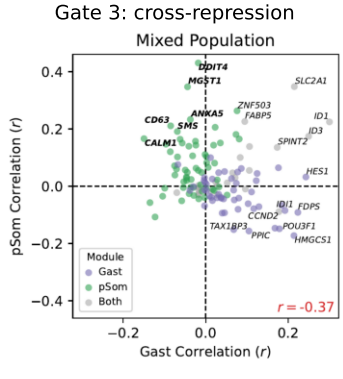


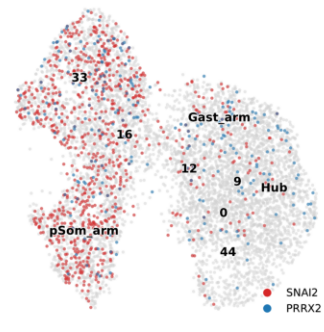












Dataset. Cutaneous SCC (Ji et al. 2020), 10x; 10531 malignant cells over 8 samples after thresholding (8 of 10 samples; clean 6.7x per-sample count gap, keep >=100). Developmental compartment 6800 cells (3731 cycling scoped out).

Verdict. ✓ **The embryonic pattern reproduces as a BRANCHING hub-and-spoke — a Gast/pSom-high hub diverging into competing Gast and pSom arms with hub cross-repression r = -0.37, the Puram signature on squamous biology.**

Parameter choices

Min-cell threshold	>=100 per sample (6.7x count gap 40->268); drops P3, P5
Overdispersed genes	1443 (scFates find_overdispersed)
Embedding	PCA 50 -> Harmony(sample, theta=5) -> kNN n_pcs=20 (scree elbow PC17), UMAP spread=1.3
Trajectory / geometry	scFates PPT tree (branching), Nodes=50 lambda=2000 sigma=0.5; dev compartment, fold-free
Root (hub)	Gast/pSom-high, Crest-low convergence node (Crest is developmentally late = terminal)
Hub identity	Hub milestone, 1195 cells; module mutual r = -0.37 (p=0.007)
Crest arm	terminal readout toward arm tips (rho_pt = +0.25); not on the cross-repression axis
EMT-TFs shown	SNAI2 (50%, late/terminal) + PRRX2 (9%, root-end); SNAI1/ZEB1 near 10x floor
vs references	hub r -0.37 ≈ Filbin(branch) -0.34; opposite Sun(linear) +0.40. pSom~Crest -0.72 (branch-like)